A proof of Seymour's second-neighborhood conjecture

Rewritten for readability from the OpenAI manuscript of the same title

September 23, 2026

Abstract

We prove that every nonempty finite oriented graph has a vertex with at least as many vertices at directed distance two as at directed distance one. This resolves Seymour's second neighborhood conjecture positively.

1 The main result

Setting. An *oriented graph* is a finite directed graph with no loops, no multiple arcs, and no pair of opposite arcs. Its underlying undirected graph need not be complete or connected. We write $v \to u$ when there is an arc from v to u. For a vertex v put

$$N_1^+(v) = \{u : v \to u\}, \qquad N_2^+(v) = \{w \notin \{v\} \cup N_1^+(v) : \text{ there is a vertex } u \text{ with } v \to u \to w\}.$$

Thus $N_2^+(v)$ is the set of vertices at directed distance exactly two from v; each such vertex is counted once, however many paths reach it.

Theorem 1.1. *Every nonempty finite oriented graph has a vertex v such that*

$$|N_1^+(v)| \le |N_2^+(v)|.$$

There is no connectivity or degree hypothesis. Two sanity checks: a sink ($N_1^+(v) = \varnothing$) satisfies the conclusion trivially, and in a directed triangle every vertex has $|N_1^+(v)| = |N_2^+(v)| = 1$.

The image operator. For a digraph D and a set $S \subseteq V(D)$ define

$$F_D(S) = \{y : \text{ there is } x \in S \text{ with } x \to y\}, \qquad F_D^2(S) = F_D(F_D(S)).$$

Note that F_D does *not* subtract S: $F_D(S)$ may meet S. We drop the subscript when the graph is clear. The following elementary facts are used throughout without further comment.

Fact 1.2. *Let D be a digraph and $A, B \subseteq V(D)$.*

(a) *$F(\varnothing) = \varnothing$, $F(A \cup B) = F(A) \cup F(B)$, and $A \subseteq B$ implies $F(A) \subseteq F(B)$.*

(b) *If D' is obtained from D by deleting arcs, then $F_{D'}(A) \subseteq F_D(A)$ and $F_{D'}^2(A) \subseteq F_D^2(A)$.*

(c) *$F(V(D)) = V(D)$ if and only if every vertex has positive indegree.*

(d) *If D is oriented, then $N_1^+(v) = F(\{v\})$, $v \notin F(\{v\}) \cup F^2(\{v\})$, and $N_2^+(v) = F^2(\{v\}) \setminus F(\{v\})$.*

Proof. (a)–(c) are immediate from the definition. For (d): $v \notin F(\{v\})$ because there are no loops, and $v \in F^2(\{v\})$ would give $v \to u \to v$, a pair of opposite arcs. By definition $N_2^+(v) = F^2(\{v\}) \setminus (\{v\} \cup F(\{v\}))$, and removing v changes nothing since $v \notin F^2(\{v\})$. □

Consequences. Section 7 derives vertex- and arc-weighted forms of the theorem through Seacrest's equivalences [19]. Section 8 records consequences for short directed cycles. For example, an oriented graph with minimum in- and outdegree at least one third of its order contains a directed triangle.

2 Proof at a glance

The proof is by contradiction. It has two halves that are logically independent and meet at one inequality. For an oriented graph G consider the *strict concavity inequality*

$$|U| + |F_G^2(U)| < 2|F_G(U)| \qquad \text{for every } \varnothing \neq U \subsetneq V(G). \tag{1}$$

- **Half A (Section 4, Proposition 4.1).** If Theorem 1.1 fails, then some nonempty oriented graph G with all indegrees positive satisfies (1).
- **Half B (Sections 5 and 6, Proposition 6.1).** No nonempty oriented graph with all indegrees positive satisfies (1).

Half A in two steps.

A1. *Subset deficit (Lemma 4.2).* Take a counterexample D with the fewest vertices and then the fewest arcs. Then every nonempty proper $S \subsetneq V(D)$ satisfies $|F^2(S) \setminus F(S)| < |F(S) \setminus S|$. Idea: delete the arcs from S into part of $F(S) \setminus S$. By minimality some vertex v of the smaller graph satisfies the second-neighborhood inequality, and v must have lost a set Q of out-neighbors. Comparing v before and after the deletion bounds the growth of a counting function by strictly less than $|Q|$; summing over a chain of such steps gives the deficit.

A2. *Blowup (Lemma 4.3).* Replace each vertex of D by a transitive tournament on $m > |V(D)|$ vertices. The integer deficit of step A1 is multiplied by m and beats an error term of at most $|V(D)|$, which yields (1).

Half B in three steps.

B1. *A rank fact (Lemma 5.1).* If a matrix has maximum rank among matrices with a prescribed zero pattern, its left and right kernels cannot both be nonzero at an allowed entry.

B2. *Pruning (Lemma 5.2).* Given two families $\mathcal{R}, \mathcal{C}$ of ordered pairs and a "conflict" relation between them, one can delete members to remove all conflicts, keeping every conflict-free member, at a cost of at most $|H|$ plus the number of points of Z left uncovered *after* the deletion (H, Z are two sets determined by the conflicts). The proof finds a small vertex cover in an auxiliary bipartite graph; by Kőnig's theorem this reduces to bounding a maximum matching by $|Z| + |H|$, which is done by comparing ranks of four matrices satisfying $NL = AB$, using B1. This lemma holds for any finite binary relation.

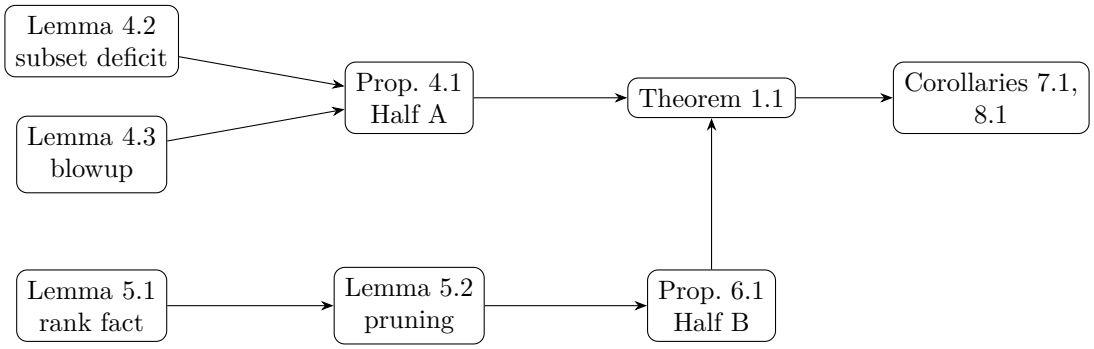
B3. *Extremal contradiction (Proposition 6.1).* Work with pairs of families $P, Q \subseteq V(G) \times V(G)$ that contain the diagonal and are *compatible* (their coordinate images are disjoint). Let $M = |P| + |Q|$ and let d be the number of points in neither image; choose (P, Q) maximizing $M + d$. Applying (1) to the columns of P and rows of Q and passing to complements of second images gives new families of total size $> M + 2d$. They may conflict; pruning (B2)

repairs this at cost at most $d + d'$, where d' is the new number of uncovered points. The new objective is therefore

$$> \; M + 2d - (d + d') + d' = M + d,$$

contradicting maximality. This is why the pruning cost must be charged partly to points left uncovered after pruning, and why d is part of the objective.

Dependencies.



Corollary 7.1 additionally uses two cited results of [19].

Where orientation is used. The absence of opposite arcs enters through Fact 1.2(d) in Lemma 4.2, in checking that the blowup is oriented, and in Proposition 6.1 at exactly one place: the diagonal families are compatible. Lemmas 5.1 and 5.2 do not use it.

Notation and renamings. Letters introduced inside a proof are local to that proof. To avoid clashes the following symbols differ from the source manuscript; nothing else is changed.

- The source's support-rank matrices D, $D^{p,i}$ and supports T, $T^{p,i}$ (Section 5) are written Φ, $\Phi^{p,i}$ and Σ, $\Sigma^{p,i}$; D is reserved for digraphs.
- Label sets inside a block of the blowup are Λ, Λ_v (source: T, T_v).
- Summation indices over R_α, C_β are ρ, γ (source: a, b, which are also names of matrix coefficients).
- A generic subset of $X \times X$ in Section 6 is Y (source: S).
- In Section 7 the sign matrix is Ψ_D (source: M_D), and in Section 8 the non-neighbor set is $N^0(v)$ (source: $Z(v)$).
- In Section 5 the letters u, v, x, y denote vectors; elements of the ground set X are called p, i, s, j there.

3 Context and prior work

This section is not needed for the proof.

The conjecture was recorded by Dean and Latka [8]. A *tournament* is an oriented graph with an arc between every pair of distinct vertices. This case, also known as Dean's conjecture,

was proved by Fisher [12] using a probability distribution obtained from Farkas' lemma. Havet and Thomassé gave a combinatorial proof using local median orders [14]. Fidler and Yuster [11] extended the tournament theorem to tournaments missing a matching, a star, or a clique. Ghazal corrected the star-case proof and treated generalized stars [13]. Subsequent advances include a matching together with a star [7] and two stars [6].

Other approaches impose degree conditions or weaken the desired ratio. Kaneko and Locke established the conjecture for minimum outdegree at most six [16]; a recent computer-assisted preprint treats minimum outdegree seven [18]. Chen, Shen and Yuster proved a universal second-to-first neighborhood ratio of approximately 0.657298 [5]. Huang and Peng raise this to approximately 0.715538 and additionally report a computational bound of 0.74530 [15]. Botler, Moura and Naia [2], followed by Espuny Díaz, Girão, Granet and Kronenberg [10], studied every orientation of binomial random graphs; the latter result covers every fixed edge probability below $1/2$, with probability tending to one as the order tends to infinity. Brukhman's recent dense-case preprint concerns graphs of order $n \le 2\delta + 2$, where δ is the minimum outdegree [4].

The present proof builds on the minimal-counterexample and graph-product methods of Brantner, Brockman, Kay and Snively [3] and on Seacrest's subset perspective [20]. The arc-deletion argument in Lemma 4.2 adapts the proof idea of [20, Lemma 4]. Specializing Seacrest's parameter λ to 1, his statement gives $|F^2(S) \setminus F(S)| < |F(S)|$ in an edge-minimal counterexample when $F(S)$ is nonempty; Lemma 4.2 replaces this right-hand bound by $|F(S) \setminus S|$ for nonempty proper S, under the stronger choice of minimum order and then minimum arc count, and is proved here in full. The blowup of Lemma 4.3 is the lexicographic product of D with a transitive tournament; the same construction appears in [3, Theorem 4.3], but here an estimate is needed for every nonempty proper subset rather than only for individual vertices. The main additional ingredient is the pruning bound of Lemma 5.2.

4 Half A: from a counterexample to strict concavity

Proposition 4.1 (Counterexample reduction)**.** *If Theorem 1.1 has a counterexample, then there is a finite nonempty oriented graph G in which every vertex has positive indegree and*

$$|U| + |F_G^2(U)| < 2|F_G(U)| \qquad \text{for every } \varnothing \neq U \subsetneq V(G). \tag{2}$$

The proof is at the end of this section; it combines the next two lemmas.

4.1 Subset deficit in a minimal counterexample

The following lemma is used once, as the input to Lemma 4.3. Throughout this subsection $F = F_D$.

Lemma 4.2 (Subset deficit)**.** *Let D be a finite nonempty oriented graph such that $|N_1^+(v)| > |N_2^+(v)|$ for every $v \in V(D)$. Choose D with minimum order and, subject to that, minimum number of arcs among all graphs with this property. Then D is strongly connected, every vertex has positive indegree, and*

$$|F^2(S) \setminus F(S)| < |F(S) \setminus S| \qquad \text{for every } \varnothing \neq S \subsetneq V(D). \tag{3}$$

Proof. Call a finite nonempty oriented graph *bad* if $|N_1^+(v)| > |N_2^+(v)|$ for all its vertices. So D is bad, no bad graph has fewer vertices, and no bad graph on $|V(D)|$ vertices has fewer arcs.

Plan. Steps 1–3 give the structural claims. For a fixed S, we write $E = F(S) \setminus S$ and define a counting function g on subsets of E with $g(\varnothing) = 0$ and $g(E) = |F^2(S) \setminus F(S)|$. Step 5 shows

that any $T \subsetneq E$ can be enlarged by a nonempty set Q with $g(T \cup Q) - g(T) < |Q|$; Steps 6–7 sum these increments.

Step 1 (D has no sink and at least two vertices). If $N_1^+(v) = \varnothing$ then $0 > |N_2^+(v)|$, which is impossible. So every vertex has an out-neighbor, which is a different vertex since there are no loops.

Step 2 (D is strongly connected). The strong components of D partition $V(D)$, and the digraph of components is acyclic, so some strong component K has no arc leaving it. For $v \in K$ we get $N_1^+(v) \subseteq K$, and every path $v \to u \to w$ lies in K. Hence $N_1^+(v)$ and $N_2^+(v)$ are the same whether computed in D or in the induced subgraph $D[K]$, so $D[K]$ is bad. By minimality of the order, $K = V(D)$.

Step 3 (positive indegrees). Given a vertex w, pick $v \neq w$ (Step 1). By Step 2 there is a directed path from v to w; its last arc enters w.

Step 4 (setup for a fixed S). Fix $\varnothing \neq S \subsetneq V(D)$ and put

$$I = F(S) \cap S, \qquad E = F(S) \setminus S, \qquad g(T) = |F(T) \setminus (I \cup T)| \quad (T \subseteq E).$$

Then $F(S) = I \sqcup E$, $I \subseteq S$, $E \cap S = \varnothing$, and $g(\varnothing) = 0$. Moreover $E \neq \varnothing$: take $x \in S$, $y \notin S$ and a directed path from x to y (Step 2); the head of its first arc leaving S lies in $F(S) \setminus S$.

Step 5 (one enlargement). *Claim: for every $T \subsetneq E$ there is a nonempty $Q \subseteq E \setminus T$ with $g(T \cup Q) - g(T) < |Q|$.*

(a) *A good vertex after deletion.* Let D_T be obtained from D by deleting all arcs $x \to y$ with $x \in S$ and $y \in E \setminus T$. Since $E \setminus T \neq \varnothing$ and every vertex of $E \subseteq F(S)$ has an in-neighbor in S, at least one arc is deleted. So D_T is an oriented graph on $V(D)$ with fewer arcs, hence not bad: there is a vertex v with

$$|N^+_{1,D_T}(v)| \leq |N^+_{2,D_T}(v)|. \tag{4}$$

(b) *v lost an out-neighbor.* Suppose $N^+_{1,D_T}(v) = N^+_{1,D}(v)$. By Fact 1.2(b),(d), $N^+_{2,D_T}(v) = F^2_{D_T}(\{v\}) \setminus N^+_{1,D}(v) \subseteq F^2_D(\{v\}) \setminus N^+_{1,D}(v) = N^+_{2,D}(v)$. Then $|N^+_{1,D}(v)| \leq |N^+_{2,D_T}(v)| \leq |N^+_{2,D}(v)|$, contradicting badness of D. So some arc out of v was deleted. Deleted arcs start in S and end in $E \setminus T$; hence $v \in S$, and the set of lost out-neighbors is exactly

$$Q = F(\{v\}) \cap (E \setminus T) \neq \varnothing, \qquad N^+_{1,D_T}(v) = N^+_{1,D}(v) \setminus Q.$$

(c) *Location of the out-neighbors of v.* Since $v \in S$, $N^+_{1,D}(v) \subseteq F(S) = I \cup E$, and its part in $E \setminus T$ is Q. Hence

$$N^+_{1,D}(v) \subseteq I \cup T \cup Q, \qquad N^+_{1,D_T}(v) \subseteq I \cup T, \qquad Q \cap (I \cup T) = \varnothing,$$

the last because $Q \subseteq E \setminus T$ and $E \cap I = \varnothing$.

(d) *Where two-step paths end after deletion.* We claim

$$F^2_{D_T}(\{v\}) \subseteq I \cup T \cup F(T). \tag{5}$$

Let $v \to u \to w$ in D_T. By (c), $u \in I \cup T$. If $u \in I \subseteq S$, then $w \in F(S) = I \cup E$, and $w \notin E \setminus T$ because all arcs from S to $E \setminus T$ were deleted; so $w \in I \cup T$. If $u \in T$, then $w \in F(T)$.

(e) *Gains.* We claim $N^+_{2,D_T}(v) \setminus N^+_{2,D}(v) \subseteq Q \cap F(T)$. Let w be in the left side. Then $w \in F^2_{D_T}(\{v\}) \subseteq F^2_D(\{v\})$. As $w \notin N^+_{2,D}(v) = F^2_D(\{v\}) \setminus N^+_{1,D}(v)$, we get $w \in N^+_{1,D}(v)$. As $w \in N^+_{2,D_T}(v)$, we have $w \notin N^+_{1,D_T}(v) = N^+_{1,D}(v) \setminus Q$. So $w \in Q$. By (5) and $Q \cap (I \cup T) = \varnothing$, also $w \in F(T)$.

(f) *Losses.* Define

$$C = F(Q) \setminus (I \cup T \cup Q \cup F(T)).$$

We claim $C \subseteq N^+_{2,D}(v)$ and $C \cap N^+_{2,D_T}(v) = \varnothing$. Each $c \in C$ has an in-neighbor $q \in Q \subseteq N^+_{1,D}(v)$, so $c \in F^2_D(\{v\})$; and $c \notin N^+_{1,D}(v)$ by (c). By Fact 1.2(d), $c \in N^+_{2,D}(v)$. (Orientation is used here: $c \neq v$ because an arc $q \to v$ would oppose $v \to q$.) The second claim holds because C is disjoint from $I \cup T \cup F(T)$, which contains $F^2_{D_T}(\{v\})$ by (5).

(g) *Counting.* Write $d_1 = |N^+_{1,D}(v)|$ and $d_2 = |N^+_{2,D}(v)|$. By (e) and (f), $N^+_{2,D_T}(v) \subseteq \big(N^+_{2,D}(v) \setminus C\big) \cup \big(Q \cap F(T)\big)$ with $C \subseteq N^+_{2,D}(v)$. Together with (b) and (4),

$$d_1 - |Q| = |N^+_{1,D_T}(v)| \leq |N^+_{2,D_T}(v)| \leq d_2 - |C| + |Q \cap F(T)|.$$

(h) *The change in g is exact.* Since $F(T \cup Q) = F(T) \cup F(Q)$,

$$F(T \cup Q) \setminus (I \cup T \cup Q) = \Big[\big(F(T) \setminus (I \cup T)\big) \setminus Q\Big] \;\sqcup\; \Big[F(Q) \setminus (I \cup T \cup Q \cup F(T))\Big].$$

The second bracket is C. In the first, the removed set is $Q \cap \big(F(T) \setminus (I \cup T)\big) = Q \cap F(T)$, because $Q \cap (I \cup T) = \varnothing$. Hence $g(T \cup Q) = g(T) - |Q \cap F(T)| + |C|$.

(i) *Conclusion of Step 5.* By (h), (g), and $d_2 < d_1$,

$$g(T \cup Q) - g(T) = |C| - |Q \cap F(T)| \leq |Q| + d_2 - d_1 < |Q|. \tag{6}$$

Step 6 (iteration). Put $T_0 = \varnothing$. While $T_k \neq E$, apply Step 5 to $T = T_k$ (each time deleting arcs afresh from the original D) to get a nonempty $Q_k \subseteq E \setminus T_k$, and set $T_{k+1} = T_k \sqcup Q_k$. The sets strictly grow inside the finite set E, so $T_r = E$ for some r, and $r \geq 1$ because $E \neq \varnothing$. Summing (6) and using $g(\varnothing) = 0$,

$$g(E) = \sum_{k<r} \big(g(T_{k+1}) - g(T_k)\big) < \sum_{k<r} |Q_k| = |E|.$$

Step 7 (identifying $g(E)$). Since $F(S) = I \cup E$ and $F(I) \subseteq F(S) = I \cup E$ (as $I \subseteq S$),

$$F^2(S) \setminus F(S) = \big(F(I) \cup F(E)\big) \setminus (I \cup E) = F(E) \setminus (I \cup E),$$

whose cardinality is $g(E)$. With $|E| = |F(S) \setminus S|$, Step 6 is (3). □

4.2 Amplifying the deficit by a blowup

The next lemma turns (3) into the strict concavity inequality; it is used only in the proof of Proposition 4.1. Each unit of the base deficit contributes a full block of size m, while the internal structure of the blocks costs at most one unit per base vertex.

Lemma 4.3 (Transitive-tournament blowup)**.** *Let D be an oriented graph on $n \geq 1$ vertices, all of positive indegree, satisfying* (3) *for every nonempty proper subset of $V(D)$. For an integer $m > n$, replace each vertex by a transitive tournament on m vertices. For every arc $v \to w$ of D, put all arcs from the block at v to the block at w, and put no other arcs between distinct blocks. The resulting graph G is oriented, all its indegrees are positive, and it satisfies* (2)*.*

Proof. Concretely, $V(G) = V(D) \times \{1, \dots, m\}$, and $(v, a) \to (w, b)$ if and only if either $v \to w$ in D (an *external* arc) or $v = w$ and $a < b$ (an *internal* arc). The *block* at v is $\{v\} \times \{1, \dots, m\}$. Here $F = F_D$.

Plan. Steps 3–5 compute $|F_G(U)|$ exactly and bound $|F_G^2(U)|$ block by block. Steps 6–7 combine them into $m \cdot (\text{base deficit}) + n$. Step 8 treats the one case in which the base hypothesis is unavailable.

Step 1 (inside one block). For a set $\Lambda \subseteq \{1, \dots, m\}$ of labels let $f(\Lambda) = \{b : a < b \text{ for some } a \in \Lambda\}$. So $f(\varnothing) = \varnothing$, and if $\Lambda \neq \varnothing$ has least element a then $f(\Lambda) = \{a+1, \dots, m\}$ and $f^2(\Lambda) = \{a+2, \dots, m\}$. Since $\Lambda \subseteq \{a, \dots, m\}$, for every Λ,

$$|\Lambda| - |f(\Lambda)| \leq 1, \qquad f^2(\Lambda) \subseteq f(\Lambda), \qquad |\Lambda| - 2|f(\Lambda)| + |f^2(\Lambda)| \leq 1. \tag{7}$$

The third is the sum of the first and $|f^2(\Lambda)| - |f(\Lambda)| \leq 0$.

Step 2 (G is oriented with positive indegrees). There are no loops. Opposite external arcs would give opposite arcs in D; opposite internal arcs would need $a < b$ and $b < a$. For a vertex (w, b), take an in-neighbor x of w in D; then $(x, 1) \to (w, b)$.

Step 3 (block rule). For any $U^* \subseteq V(G)$ let $\Lambda_w^* = \{a : (w, a) \in U^*\}$ and $S^* = \{w : \Lambda_w^* \neq \varnothing\}$. Then for every $w \in V(D)$:

$$F_G(U^*) \cap (\text{block at } w) = \begin{cases} \text{the whole block}, & w \in F(S^*), \\ \{w\} \times f(\Lambda_w^*), & w \notin F(S^*). \end{cases}$$

Indeed, (w, b) has an in-neighbor in U^* through an external arc iff $w \in F(S^*)$ (independently of b), and through an internal arc iff $b \in f(\Lambda_w^*)$.

Step 4 (first image). Fix $\varnothing \neq U \subsetneq V(G)$, and write $\Lambda_v = \{a : (v, a) \in U\}$,

$$S = \{v \in V(D) : \Lambda_v \neq \varnothing\} \neq \varnothing, \qquad H_0 = F(S) \cup F^2(S).$$

By Step 3 with $U^* = U$ (blocks outside $S \cup F(S)$ contribute $f(\varnothing) = \varnothing$),

$$|F_G(U)| = m|F(S)| + \sum_{v \in S \setminus F(S)} |f(\Lambda_v)|.$$

Step 5 (second image). Apply Step 3 with $U^* = F_G(U)$. By Step 4 its support S^* is contained in $F(S) \cup S$, so $F(S^*) \subseteq F^2(S) \cup F(S) = H_0$. Let $w \notin H_0$. Then $w \notin F(S^*)$, and since $w \notin F(S)$ the label set of $F_G(U)$ in the block at w is $f(\Lambda_w)$ (Step 4). The block rule gives $F_G^2(U) \cap (\text{block at } w) = \{w\} \times f^2(\Lambda_w)$, which is empty unless $w \in S$. Blocks indexed by H_0 contribute at most m each. Therefore

$$|F_G^2(U)| \leq m|H_0| + \sum_{v \in S \setminus H_0} |f^2(\Lambda_v)|.$$

Step 6 (combination). Using $|U| = \sum_{v \in S} |\Lambda_v|$ and Steps 4–5,

$$\begin{aligned} |U| - 2|F_G(U)| + |F_G^2(U)| \leq m(|H_0| - 2|F(S)|) &+ \sum_{v \in S} |\Lambda_v| \\ &- 2 \sum_{v \in S \setminus F(S)} |f(\Lambda_v)| + \sum_{v \in S \setminus H_0} |f^2(\Lambda_v)|. \end{aligned} \tag{8}$$

Since $F(S) \subseteq H_0$, we have $S \setminus H_0 \subseteq S \setminus F(S)$, so extending the last sum to $S \setminus F(S)$ only adds nonnegative terms. Split $\sum_{v\in S}|\Lambda_v|$ over $S \cap F(S)$ and $S \setminus F(S)$; use $|\Lambda_v| \le m$ on the first part and the third inequality of (7) on the second:

$$\begin{aligned}|U| - 2|F_G(U)| + |F_G^2(U)| &\le m\big(|H_0| - 2|F(S)|\big) + \sum_{v\in S\cap F(S)} |\Lambda_v| + \sum_{v\in S\setminus F(S)} \big(|\Lambda_v| - 2|f(\Lambda_v)| + |f^2(\Lambda_v)|\big)\\ &\le m\big(|H_0| - 2|F(S)| + |S \cap F(S)|\big) + |S \setminus F(S)|\\ &\le m\big(|F^2(S) \setminus F(S)| - |F(S) \setminus S|\big) + n.\end{aligned}$$

The last line uses $|H_0| = |F(S)|+|F^2(S)\setminus F(S)|$, $|F(S)|-|S\cap F(S)| = |F(S)\setminus S|$, and $|S\setminus F(S)| \le n$.

Step 7 (case $S \subsetneq V(D)$). Then S is nonempty and proper, so by (3) the parenthesized difference is an integer ≤ -1. The bound of Step 6 is at most $-m + n < 0$, which is (2) for U.

Step 8 (case $S = V(D)$). Here (3) is not available. Positive indegrees in D give $F(S) = V(D)$, so by Step 3 every block is filled: $F_G(U) = V(G)$. Positive indegrees in G (Step 2) give $F_G^2(U) = F_G(V(G)) = V(G)$. Hence

$$|U| - 2|F_G(U)| + |F_G^2(U)| = |U| - |V(G)| < 0,$$

because U is proper. □

Proof of Proposition 4.1. A counterexample to Theorem 1.1 is a finite nonempty oriented graph with $|N_1^+(v)| > |N_2^+(v)|$ for every vertex. Choose one, D, of minimum order and then minimum number of arcs. Lemma 4.2 gives positive indegrees and (3). Lemma 4.3 with any integer $m > |V(D)|$ then gives the required graph G, which is nonempty. □

5 Half B, part 1: pruning two families of pairs

This section is self-contained and involves no orientation assumption. In it, $\mathbb{R}^Y$ denotes the real vector space with standard basis $(e_y)_{y\in Y}$ for a finite set Y. A matrix $\mathbb{R}^J \to \mathbb{R}^I$ has rows indexed by I and columns by J. An empty determinant equals 1. Recall that u, v, x, y denote vectors in this section.

5.1 A rank fact

For finite index sets I, J and $\Sigma \subseteq I \times J$, a matrix *supported on* Σ is a real $I \times J$ matrix whose entries outside Σ are zero; entries in Σ may also vanish. The next lemma is used once, in Step 9 of the proof of Lemma 5.2.

Lemma 5.1 (Kernels at maximum support rank)**.** *Let $\Sigma \subseteq I \times J$, where I and J are finite. Suppose that $\Phi : \mathbb{R}^J \to \mathbb{R}^I$ has maximum rank among all real matrices supported on Σ. Then, for every $x \in \ker \Phi$, every $y \in \ker \Phi^{\mathsf{T}}$, and every $(s, j) \in \Sigma$, one has $x_j y_s = 0$.*

Proof. Suppose $x_j y_s \neq 0$ for some $(s, j) \in \Sigma$. Fix a real $\tau \neq 0$ and put $\Phi_\tau = \Phi + \tau e_s e_j^{\mathsf{T}}$, which is still supported on Σ.

(1) $\Phi_\tau x = \Phi x + \tau x_j e_s = \tau x_j e_s$, so $e_s \in \operatorname{im} \Phi_\tau$.

(2) For every $w \in \mathbb{R}^J$, $\Phi w = \Phi_\tau w - \tau w_j e_s \in \operatorname{im} \Phi_\tau$ by (1). So $\operatorname{im} \Phi \subseteq \operatorname{im} \Phi_\tau$.

(3) $y^{\mathsf{T}} \Phi w = (\Phi^{\mathsf{T}} y)^{\mathsf{T}} w = 0$ for all w, while $y^{\mathsf{T}} e_s = y_s \neq 0$. So $e_s \notin \operatorname{im} \Phi$.

By (1)–(3), $\operatorname{im} \Phi \subsetneq \operatorname{im} \Phi_\tau$, so $\operatorname{rank} \Phi_\tau > \operatorname{rank} \Phi$, contradicting maximality. □

5.2 The pruning lemma

The following lemma is used once, in Step 9 of the proof of Proposition 6.1.

Lemma 5.2 (Pruning)**.** *Let X be a finite set with a binary relation denoted by $\to$, and let $\mathcal{R}, \mathcal{C} \subseteq X \times X$ be two families, regarded as separate copies even when their coordinates agree. Say that $(p, j) \in \mathcal{R}$ and $(i, s) \in \mathcal{C}$* conflict *if*

$$p \to i \qquad \text{and} \qquad s \to j.$$

Define

$$Z = \{(i, j) : \text{some } (p, j) \in \mathcal{R} \text{ conflicts with some } (i, s) \in \mathcal{C}\},$$
$$H = \{(p, s) : \text{some } (p, j) \in \mathcal{R} \text{ conflicts with some } (i, s) \in \mathcal{C}\}.$$

A point (i, j) is covered *by a member $(p, j) \in \mathcal{R}$ when $p \to i$, and by a member $(i, s) \in \mathcal{C}$ when $s \to j$. These are the only ways a member covers a point.*

There are subfamilies $\mathcal{R}' \subseteq \mathcal{R}$ and $\mathcal{C}' \subseteq \mathcal{C}$ with no conflicts between them such that every initially conflict-free member is retained and

$$|\mathcal{R} \setminus \mathcal{R}'| + |\mathcal{C} \setminus \mathcal{C}'| \leq |H| + |\{z \in Z : z \text{ is covered by no member of } \mathcal{R}' \cup \mathcal{C}'\}|. \tag{9}$$

The two sides are counted separately throughout.

Example 5.3. Let $X = \{p, s, i_1, i_2, j_1, j_2\}$ with $p \to i_1, i_2$ and $s \to j_1, j_2$, $\mathcal{R} = \{(p, j_1), (p, j_2)\}$, $\mathcal{C} = \{(i_1, s), (i_2, s)\}$. All four pairs conflict, $H = \{(p, s)\}$ and $Z = \{i_1, i_2\} \times \{j_1, j_2\}$. Any conflict-free pruning deletes all of $\mathcal{R}$ or all of $\mathcal{C}$, so at least $2 > |H|$ members: the second term in (9) is needed. Deleting only $\mathcal{R}$ leaves all of Z covered by $\mathcal{C}$ and violates (9) ($2 \not\leq 1 + 0$); deleting everything satisfies it ($4 \leq 1 + 4$). The lemma asserts the existence of *some* pruning, not that a cheapest one works.

Proof of Lemma 5.2. If there are no conflicts, retain every member; then $Z = H = \varnothing$ and (9) reads $0 \leq 0$. Assume there are conflicts.

Plan. Steps 1–2 translate the lemma into finding a vertex cover of size at most $|Z| + |H|$ in a bipartite graph Γ. Step 3 (Kőnig) reduces this to showing that a maximum matching has at most $|Z| + |H|$ edges. Steps 4–10 prove this matching bound by linear algebra.

Step 1 (the augmented graph Γ). Let Z_L, Z_R be two disjoint copies of Z, disjoint from the families; for $z \in Z$ write z_L, z_R for its copies. Γ is the bipartite graph with sides

$$\mathcal{V}_L = \mathcal{R} \sqcup Z_L, \qquad \mathcal{V}_R = \mathcal{C} \sqcup Z_R,$$

and three kinds of edges (and no others):

(E1) $r \in \mathcal{R}$ and $c \in \mathcal{C}$ are joined when they conflict;

(E2) $r = (p, j) \in \mathcal{R}$ is joined to z_R, $z = (i, j) \in Z$, when r covers z (that is, $p \to i$);

(E3) $c = (i, s) \in \mathcal{C}$ is joined to z_L, $z = (i, j) \in Z$, when c covers z (that is, $s \to j$).

Figure 1 shows the three edges forced by one conflict.

Claim: an initially conflict-free member is isolated in Γ. It has no (E1) edge. Suppose $(p, j) \in \mathcal{R}$ covers $(i, j) \in Z$, so $p \to i$. Since $(i, j) \in Z$, some $(i, s) \in \mathcal{C}$ has $s \to j$ (it conflicts with

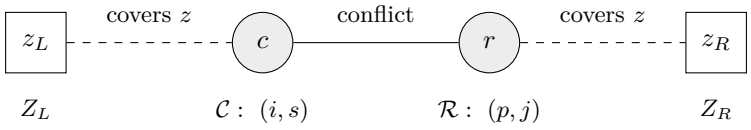


Figure 1: A conflict between $r = (p, j)$ and $c = (i, s)$ forces this path, where $z = (i, j) \in Z$. The copies z_L and z_R are distinct vertices. The vertices alternate between the two sides of the bipartition. Different conflicts can share vertices and edges; the full augmented graph need not be a disjoint union of such paths.

some member (p', j)). Then (p, j) and (i, s) conflict. So a conflict-free member of $\mathcal{R}$ has no (E2) edge. The argument for $\mathcal{C}$ and (E3) is symmetric: if $(i, s) \in \mathcal{C}$ covers $(i, j) \in Z$, then $s \to j$, and some $(p, j) \in \mathcal{R}$ has $p \to i$, so they conflict.

Step 2 (a small vertex cover suffices). *Suppose K is a vertex cover of Γ with $|K| \le |Z| + |H|$ that contains no isolated vertex. Then $\mathcal{R}' = \mathcal{R} \setminus K$, $\mathcal{C}' = \mathcal{C} \setminus K$ satisfy the lemma.*

No conflict survives, since each conflict is an (E1) edge and has an endpoint in K. Conflict-free members are isolated (Step 1), so they are retained. For the count, let q be the number of points of Z covered by no member of $\mathcal{R}' \cup \mathcal{C}'$. Each of the other $|Z| - q$ points z is covered by a surviving member; this member is joined by (E2) or (E3) to a copy of z, and since the member is not in K, that copy is in K. Copies of distinct points are distinct vertices, so $|K \cap (Z_L \sqcup Z_R)| \ge |Z| - q$. The deleted members are $K \cap (\mathcal{R} \sqcup \mathcal{C})$, whence

$$|K \cap (\mathcal{R} \sqcup \mathcal{C})| \le |K| - (|Z| - q) \le |H| + q, \tag{10}$$

which is (9).

Step 3 (Kőnig's theorem [17]: a cover from a maximum matching). *For every maximum matching $\mathcal{M}$ of Γ there is a vertex cover K with $|K| = |\mathcal{M}|$ containing no isolated vertex.* For completeness: starting at all $\mathcal{M}$-unmatched vertices of $\mathcal{V}_L$, follow edges outside $\mathcal{M}$ from left to right and edges of $\mathcal{M}$ from right to left. Let W_L, W_R be the sets of reached vertices on the two sides, and put $K = (\mathcal{V}_L \setminus W_L) \cup W_R$.

(i) No unmatched right vertex is reached; an alternating path to one would augment $\mathcal{M}$.

(ii) The endpoints of a matched edge are both reached or both unreached: a reached right endpoint leads to its mate, and a matched left vertex is not a starting vertex, so it can only be reached through its mate.

(iii) K covers every edge. An uncovered edge would go from a reached left vertex to an unreached right vertex. If unmatched, it would have been traversed; if matched, it contradicts (ii).

(iv) Every vertex of K is matched (unmatched left vertices lie in W_L; use (i) for W_R), and by (ii) each matched edge has exactly one endpoint in K. So $|K| = |\mathcal{M}|$, and K contains no isolated vertex.

By Steps 2 and 3 it remains to exhibit one maximum matching of size at most $|Z| + |H|$.

Step 4 (a special maximum matching). Choose a maximum matching $\mathcal{M}$ with the fewest (E1) edges. Let k be the number of its (E1) edges, α the set of its (E2) edges and β the set of its (E3) edges, and put

$$\ell = |\alpha|, \qquad t = |\beta|, \qquad |\mathcal{M}| = k + \ell + t.$$

Let $R_\alpha \subseteq \mathcal{R}$, $Z_\alpha \subseteq Z$ be the members and the points (via right copies) matched by α, and $C_\beta \subseteq \mathcal{C}$, $Z_\beta \subseteq Z$ those matched by β.

Claim: α is a maximum matching of the subgraph of (E2) *edges, and β of the subgraph of* (E3) *edges.* Suppose the (E2) subgraph had a larger matching. Its symmetric difference with α contains an α-augmenting path π from an α-unmatched $r_0 \in \mathcal{R}$ to an α-unmatched vertex of Z_R. Vertices of Z_R have only (E2) edges, so the end of π is unmatched in $\mathcal{M}$. Every internal vertex of π is matched in $\mathcal{M}$ by an α-edge of π. If r_0 is unmatched in $\mathcal{M}$, switching along π gives a larger matching of Γ. Otherwise r_0 lies on an (E1) edge of $\mathcal{M}$. Remove that edge and switch along π: the size is unchanged and k drops by one. Both outcomes contradict the choice of $\mathcal{M}$. The proof for β is the same with the sides exchanged.

Step 5 (four matrices with $NL = AB$). For each pair with $p \to i$ let a_{pi} be a real number, and for each pair with $s \to j$ let b_{sj} be a real number; set $a_{pi} = 0$ when $p \not\to i$ and $b_{sj} = 0$ when $s \not\to j$. (The values are chosen in Step 7; this step holds for any values.) Define

$$\begin{array}{ccc} \mathbb{R}^{\mathcal{R}} & \xrightarrow{L} & \mathbb{R}^{Z} \\ {\scriptstyle B}\downarrow & & \downarrow{\scriptstyle N} \\ \mathbb{R}^{H} & \xrightarrow{A} & \mathbb{R}^{\mathcal{C}} \end{array}$$

by

$$\begin{aligned} L_{(i,j),(p,j)} &= a_{pi}, \quad N_{(i,s),(i,j)} = b_{sj}, \\ B_{(p,s),(p,j)} &= b_{sj}, \quad A_{(i,s),(p,s)} = a_{pi}, \end{aligned} \tag{11}$$

where the indices range over the indicated coordinate sets and every entry not of a listed form is zero. For instance, the entry of L in row $(i, j') \in Z$ and column $(p, j) \in \mathcal{R}$ is a_{pi} if $j' = j$ and 0 otherwise. Thus L, A move the first coordinate forward along $p \to i$, and N, B move the second coordinate backward along $s \to j$. Two observations:

- An entry of L at row z, column r can be nonzero only if r covers z, i.e. only at (E2) edges. An entry of N^{T} at row z, column c can be nonzero only if c covers z, i.e. only at (E3) edges.
- $NL = AB$. Fix a column $(p, j) \in \mathcal{R}$ and a row $(i, s) \in \mathcal{C}$. In $(NL)_{(i,s),(p,j)} = \sum_{z \in Z} N_{(i,s),z} L_{z,(p,j)}$ the only possibly nonzero term is $z = (i, j)$, and in $(AB)_{(i,s),(p,j)}$ the only one is the index (p, s). If (p, j) and (i, s) conflict, then $(i, j) \in Z$ and $(p, s) \in H$, and both entries equal $a_{pi} b_{sj}$. If they do not conflict, then $a_{pi} = 0$ or $b_{sj} = 0$, and both entries are 0. Hence

$$NL = AB. \tag{12}$$

Step 6 (three rank conditions imply the matching bound). Suppose the coefficients are such that

$$\operatorname{rank} L = \ell, \qquad \operatorname{rank} N = t, \qquad \dim \ker A \geq k.$$

Restricting N to $\operatorname{im} L$ gives

$$\operatorname{rank}(NL) = \dim \operatorname{im} L - \dim(\operatorname{im} L \cap \ker N) \geq \ell - \dim \ker N = \ell - (|Z| - t).$$

By rank–nullity for A, $\operatorname{rank} A \geq \operatorname{rank}(AB)$, and (12),

$$\begin{aligned} |H| &= \dim \ker A + \operatorname{rank} A \\ &\geq k + \operatorname{rank} A \geq k + \operatorname{rank}(NL) \\ &\geq k + \ell + t - |Z|, \end{aligned} \tag{13}$$

that is, $|\mathcal{M}| = k + \ell + t \le |Z| + |H|$. Steps 7–10 produce such coefficients.

Step 7 (choosing the coefficients). Regard the a_{pi} $(p \to i)$ and b_{sj} $(s \to j)$ as independent indeterminates. This is the correspondence between generic matrix rank and maximum matching size [9, Section 5, Theorem 1], applied within coordinate blocks; the needed argument is given in full. For row and column index sets I', J' write $L[I', J']$ for the corresponding submatrix.

(a) *For every real assignment,* $\operatorname{rank} L \le \ell$ *and* $\operatorname{rank} N \le t$. If an $m' \times m'$ submatrix of L has nonzero determinant, some term of the Leibniz expansion is nonzero. This gives a bijection from its columns to its rows along nonzero entries, i.e. (Step 5) along (E2) edges: a matching of size m' in the (E2) subgraph. By Step 4, $m' \le \ell$. The same argument for N^{T} and (E3) gives $\operatorname{rank} N = \operatorname{rank} N^{\mathsf{T}} \le t$.

(b) *The polynomial* $\pi_\alpha = \det L[Z_\alpha, R_\alpha]$ *is nonzero.* For $j \in X$ let R_α^j, Z_α^j be the elements of R_α, Z_α with second coordinate j. Every α-edge joins some (p, j) to the copy of some (i, j) with the same j, so α restricts to a bijection $R_\alpha^j \to Z_\alpha^j$. Entries of L between different second coordinates vanish. So, after ordering rows and columns by j, $L[Z_\alpha, R_\alpha]$ is block diagonal with square blocks $L[Z_\alpha^j, R_\alpha^j]$, and π_α is $\pm$ the product of their determinants. In the block for j, the entry at row (i, j), column (p, j) is the indeterminate a_{pi} if $p \to i$ and 0 otherwise, so distinct positions carry distinct indeterminates. In the Leibniz expansion of the block determinant, each bijection all of whose positions are allowed contributes $\pm$ the product of the indeterminates at its positions. Different bijections use different sets of positions, hence give different monomials, and nothing cancels. The bijection given by α is of this kind (its edges are (E2) edges). So each block determinant is a nonzero polynomial. A variable a_{pi} may occur in several blocks, but a product of nonzero polynomials is nonzero.

(c) *The polynomial* $\pi_\beta = \det N^{\mathsf{T}}[Z_\beta, C_\beta]$ *is nonzero.* Same argument with blocks of fixed first coordinate i: the entry at row (i, j), column (i, s) is b_{sj}, and within a block distinct positions carry distinct indeterminates.

(d) *Support-rank polynomials.* For every pair with $p \to i$, define

$$J_p = \{j : (p, j) \in \mathcal{R}\}, \qquad S_i = \{s : (i, s) \in \mathcal{C}\}, \qquad \Phi^{p,i} = (b_{sj})_{s \in S_i,\ j \in J_p},$$

$$\Sigma^{p,i} = \{(s, j) \in S_i \times J_p : s \to j\}.$$

$\Phi^{p,i}$ is supported on $\Sigma^{p,i}$, and its entries on $\Sigma^{p,i}$ are distinct indeterminates, so every real matrix supported on $\Sigma^{p,i}$ is a specialization of it. Let $\varrho^{p,i}$ be the maximum rank of a real matrix supported on $\Sigma^{p,i}$, and pick such a matrix together with a nonzero $\varrho^{p,i} \times \varrho^{p,i}$ minor of it (the empty minor if $\varrho^{p,i} = 0$). The same minor of $\Phi^{p,i}$ is a polynomial $\pi^{p,i}$ in the b-variables that is nonzero at one assignment, hence nonzero.

(e) *One assignment for everything.* The product $\pi_\alpha \, \pi_\beta \prod_{p \to i} \pi^{p,i}$ is a nonzero polynomial in all the a- and b-variables, even though variables are shared between factors. A nonzero real polynomial is nonzero at some real point (induction on the number of variables). *Fix such an assignment once and for all.* By (a), (b), (c), (d), at this assignment:

- (R1) $\operatorname{rank} L = \ell$, and the columns Le_ρ, $\rho \in R_\alpha$, form a basis of $\operatorname{im} L$ (they are ℓ independent vectors in a space of dimension ℓ);
- (R2) $\operatorname{rank} N = \operatorname{rank} N^{\mathsf{T}} = t$, and the columns $N^{\mathsf{T}} e_\gamma$, $\gamma \in C_\beta$, form a basis of $\operatorname{im} N^{\mathsf{T}}$;
- (R3) for every $p \to i$, $\Phi^{p,i}$ has rank $\varrho^{p,i}$, the maximum among real matrices supported on $\Sigma^{p,i}$.

The first two rank conditions of Step 6 hold. It remains to prove $\dim\ker A \geq k$.

Step 8 (lifting the k matched conflicts into kernels). Let (r, c) be an (E1) edge of $\mathcal{M}$. Since $\mathcal{M}$ is a matching, $r \notin R_\alpha$ and $c \notin C_\beta$. By (R1), $Le_r \in \operatorname{im} L$ is a combination of the Le_ρ, and by (R2) likewise for $N^{\mathsf{T}}e_c$. So there are reals λ^r_ρ, μ^c_γ with

$$\begin{aligned} u^r &= e_r + \sum_{\rho\in R_\alpha} \lambda^r_\rho e_\rho \in \ker L, \\ v^c &= e_c + \sum_{\gamma\in C_\beta} \mu^c_\gamma e_\gamma \in \ker N^{\mathsf{T}}. \end{aligned} \tag{14}$$

Let $r_1, \dots, r_k$ be the left endpoints of the (E1) edges of $\mathcal{M}$. None lies in R_α, so the coordinate of u^{r_a} at r_b is 1 if $a = b$ and 0 otherwise. Also $v^c_c = 1$.

Step 9 (kernel vectors cannot both be nonzero on a conflict). *Claim: at the fixed assignment,*

$$u_{pj}v_{is} = 0 \quad \text{for every conflict } ((p,j),(i,s)), \quad \text{every } u \in \ker B, \text{ every } v \in \ker N^{\mathsf{T}}. \tag{15}$$

Fix such a conflict, so $p \to i$, $s \to j$, $j \in J_p$, $s \in S_i$, and $(s,j) \in \Sigma^{p,i}$. Fix $u \in \ker B$, $v \in \ker N^{\mathsf{T}}$.

(a) $(u_{pj'})_{j'\in J_p} \in \ker \Phi^{p,i}$. We must show, for every $s' \in S_i$,

$$\sum_{j'\in J_p} b_{s'j'}u_{pj'} = 0. \tag{16}$$

If $(p,s') \in H$, the left side is the (p,s')-coordinate of Bu (the columns of B meeting row (p,s') are (p,j'), $j' \in J_p$, with entries $b_{s'j'}$), which is 0. If $(p,s') \notin H$, then $s' \not\to j'$ for every $j' \in J_p$; otherwise $(p,j') \in \mathcal{R}$ and $(i,s') \in \mathcal{C}$ would conflict (as $p \to i$) and put (p,s') in H. So all coefficients vanish.

(b) $(v_{is'})_{s'\in S_i} \in \ker(\Phi^{p,i})^{\mathsf{T}}$. We must show, for every $j' \in J_p$,

$$\sum_{s'\in S_i} b_{s'j'}v_{is'} = 0. \tag{17}$$

If $(i,j') \in Z$, the left side is the (i,j')-coordinate of $N^{\mathsf{T}}v$, which is 0. If $(i,j') \notin Z$, then $s' \not\to j'$ for every $s' \in S_i$ (otherwise (p,j') and (i,s') would conflict and put (i,j') in Z), so all coefficients vanish.

(c) By (R3), Lemma 5.1 applies to $\Phi = \Phi^{p,i}$, $\Sigma = \Sigma^{p,i}$ with the vectors of (a), (b) and the entry $(s,j) \in \Sigma^{p,i}$. It gives $u_{pj}v_{is} = 0$. The perturbation inside that lemma is hypothetical and concerns only $\Phi^{p,i}$; the conclusion is about the fixed assignment.

Step 10 ($\dim\ker A \geq k$, and conclusion). Let (r,c) be an (E1) edge of $\mathcal{M}$; it is a conflict. By (15) with $v = v^c$ and $v^c_c = 1$, *every* $u \in \ker B$ has $u_r = 0$. Now suppose $\sum_a \theta_a Bu^{r_a} = 0$. Then $u = \sum_a \theta_a u^{r_a} \in \ker B$, and its coordinate at r_b is θ_b by Step 8; so all $\theta_b = 0$. Thus $Bu^{r_1}, \dots, Bu^{r_k}$ are linearly independent. They lie in $\ker A$, because by (12) and (14)

$$A(Bu^r) = N(Lu^r) = 0.$$

Hence $\dim\ker A \geq k$. By Step 6, $|\mathcal{M}| \leq |Z| + |H|$; by Step 3 there is a vertex cover K with $|K| = |\mathcal{M}|$ and no isolated vertex; by Step 2 the lemma follows. □

6 Half B, part 2: the extremal contradiction

Proposition 6.1. *There is no nonempty finite oriented graph G in which every vertex has an in-neighbor and whose one-step image operator*

$$F(U) = \{y \in V(G) : x \to y \text{ for some } x \in U\}$$

satisfies

$$|U| + |F^2(U)| < 2|F(U)| \qquad (\varnothing \neq U \subsetneq V(G)), \tag{18}$$

where $F^2 = F \circ F$.

Proof. Suppose such a G exists and put $X = V(G)$, with $\to$ the arc relation.

Plan. Steps 1–3 set up an extremal pair (P, Q). Steps 4–5 build larger families $\mathcal{R}, \mathcal{C}$. Steps 6–8 verify the hypotheses needed to prune them with Lemma 5.2. Step 9 contradicts maximality.

Step 1 (coordinate images and compatibility). For $Y \subseteq X \times X$ define

$$F_1 Y = \{(i, j) : (p, j) \in Y \text{ and } p \to i \text{ for some } p \in X\},$$
$$F_2 Y = \{(i, j) : (i, s) \in Y \text{ and } s \to j \text{ for some } s \in X\}.$$

Write $F_1^2 = F_1 \circ F_1$, $F_2^2 = F_2 \circ F_2$, and $\Delta = \{(v, v) : v \in X\}$. Both operators are monotone. For the column $Y_j = \{p : (p, j) \in Y\}$, the column of $F_1 Y$ at j is $F(Y_j)$ and that of $F_1^2 Y$ is $F^2(Y_j)$. Likewise, for the row $Y^i = \{s : (i, s) \in Y\}$, the rows of $F_2 Y$, $F_2^2 Y$ at i are $F(Y^i)$, $F^2(Y^i)$.

Call $P, Q \subseteq X \times X$ *compatible* if $F_1 P \cap F_2 Q = \varnothing$. Unwinding the definitions, this says that no $(p, j) \in P$ and $(i, s) \in Q$ satisfy both $p \to i$ and $s \to j$. That is, no member of P (as left family) conflicts with a member of Q (as right family) in the sense of Lemma 5.2.

Step 2 (an extremal pair). (Δ, Δ) is compatible: a point $(i, j) \in F_1\Delta \cap F_2\Delta$ would require $j \to i$ and $i \to j$, contrary to orientation. So, X being finite, we may choose among all compatible pairs with $\Delta \subseteq P \cap Q$ one maximizing $M + d$, where

$$M = |P| + |Q|, \qquad d = |X|^2 - |F_1 P| - |F_2 Q|.$$

By compatibility, d is the number of points outside $F_1 P \cup F_2 Q$ (the *uncovered* points).

Step 3 (columns of P and rows of Q are nonempty and proper). The column P_j contains j. Suppose $P_j = X$. Positive indegrees give $F(X) = X$, so $X \times \{j\} \subseteq F_1 P$. Choose an in-neighbor w of j; then $(w, j) \in F_2\Delta \subseteq F_2 Q$, contradicting compatibility. Likewise the row Q^i contains i, and if $Q^i = X$ then $\{i\} \times X \subseteq F_2 Q$, while $w \to i$ gives $(i, w) \in F_1\Delta \subseteq F_1 P$, again a contradiction.

Step 4 (summing the concavity inequality). Apply (18) to $U = P_j$ for each $j \in X$ and sum, using Step 1; do the same for the rows Q^i. Since $X \neq \varnothing$ the sums are strict:

$$|P| + |F_1^2 P| < 2|F_1 P|, \qquad |Q| + |F_2^2 Q| < 2|F_2 Q|. \tag{19}$$

Step 5 (new families and their size). Define

$$\mathcal{R} = (X \times X) \setminus F_2^2 Q, \qquad \mathcal{C} = (X \times X) \setminus F_1^2 P.$$

By (19),

$$\begin{aligned} |\mathcal{R}| + |\mathcal{C}| &= 2|X|^2 - |F_2^2 Q| - |F_1^2 P| \\ &> 2|X|^2 + M - 2\big(|F_1 P| + |F_2 Q|\big) = M + 2d. \end{aligned} \tag{20}$$

These families need not be compatible.

Step 6 (both new families contain Δ). If $(v,v) \in F_2^2 Q$, there is $s \to v$ with $(v,s) \in F_2 Q$. But $(s,s) \in P$ and $s \to v$ give $(v,s) \in F_1 P$, contradicting compatibility of (P,Q). So $\Delta \subseteq \mathcal{R}$. If $(v,v) \in F_1^2 P$, there is $p \to v$ with $(p,v) \in F_1 P$; and $(p,p) \in Q$, $p \to v$ give $(p,v) \in F_2 Q$, the same contradiction. So $\Delta \subseteq \mathcal{C}$.

Step 7 (diagonal members are conflict-free). Suppose the left member $(v,v) \in \mathcal{R}$ conflicts with a right member $(i,s) \in \mathcal{C}$: $v \to i$ and $s \to v$. From $(s,s) \in P$ we get $(v,s) \in F_1 P$ and then $(i,s) \in F_1^2 P$, contradicting $(i,s) \in \mathcal{C}$. Suppose a left member $(p,j) \in \mathcal{R}$ conflicts with the right member $(v,v) \in \mathcal{C}$: $p \to v$ and $v \to j$. From $(p,p) \in Q$ we get $(p,v) \in F_2 Q$ and then $(p,j) \in F_2^2 Q$, contradicting $(p,j) \in \mathcal{R}$.

Step 8 ($|H| \le d$). Let H, Z be the sets attached to $\mathcal{R}, \mathcal{C}$ in Lemma 5.2. We claim

$$H \subseteq (X \times X) \setminus (F_1 P \cup F_2 Q), \qquad \text{and hence} \qquad |H| \le d. \tag{21}$$

Let $(p,s) \in H$, witnessed by a conflict $(p,j) \in \mathcal{R}$, $(i,s) \in \mathcal{C}$, $p \to i$, $s \to j$. If $(p,s) \in F_1 P$, choose $(x,s) \in P$ with $x \to p$; the path $x \to p \to i$ puts (i,s) in $F_1^2 P$, contrary to $(i,s) \in \mathcal{C}$. If $(p,s) \in F_2 Q$, choose $(p,h) \in Q$ with $h \to s$; the path $h \to s \to j$ puts (p,j) in $F_2^2 Q$, contrary to $(p,j) \in \mathcal{R}$. The bound $|H| \le d$ follows from Step 2.

Step 9 (pruning and the contradiction). Apply Lemma 5.2 to $\mathcal{R}, \mathcal{C}$, obtaining conflict-free $P' \subseteq \mathcal{R}$ and $Q' \subseteq \mathcal{C}$.

- By Step 7 and the retention clause, $\Delta \subseteq P' \cap Q'$.
- By Step 1, conflict-freeness means $F_1 P' \cap F_2 Q' = \varnothing$. So (P', Q') is a competitor in Step 2, and
$$d' = |X|^2 - |F_1 P'| - |F_2 Q'|$$
is its number of uncovered points.
- A point is covered (in the sense of the lemma) by a kept left member exactly when it lies in $F_1 P'$, and by a kept right member exactly when it lies in $F_2 Q'$. So at most d' points of Z are uncovered after pruning.
- By (9) and (21), the number of deleted members is at most $|H| + d' \le d + d'$.

Using (20),

$$\begin{aligned} |P'| + |Q'| + d' &\ge |\mathcal{R}| + |\mathcal{C}| - (d + d') + d' \\ &> M + 2d - (d + d') + d' = M + d. \end{aligned}$$

This contradicts the maximality of $M + d$. □

Proof of Theorem 1.1. If the theorem were false, Proposition 4.1 would produce a nonempty finite oriented graph with positive indegrees satisfying (2), which is the same as (18), for every nonempty proper vertex set. Proposition 6.1 rules out such a graph. □

7 Weighted consequences

The following corollary applies Theorem 1.1 through Seacrest's established equivalences of the corresponding assertions over all nonempty finite oriented graphs [19]. The equivalences themselves are not new here. The symbol β_v^w below is unrelated to the matching β of Section 5.

Corollary 7.1 (Weighted second-neighborhood forms)**.** *Let D be a nonempty finite oriented graph, and write $A(D)$ for its arc set. For a vertex weighting $\eta : V(D) \to [0,\infty)$, write $\eta(S) = \sum_{x\in S} \eta(x)$. Then the following hold.*

(i) *For every vertex weighting $\eta : V(D) \to [0,\infty)$, there is a vertex $v \in V(D)$ such that*

$$\eta(N_1^+(v)) \le \eta(N_2^+(v)).$$

(ii) *There is a vertex weighting $\eta : V(D) \to [0,\infty)$ with $\sum_{x\in V(D)} \eta(x) = 1$ such that*

$$\eta(N_1^+(v)) \le \eta(N_2^+(v)) \qquad \text{for every } v \in V(D).$$

The entries of this weighting need not all be positive.

(iii) *For every arc weighting $w : A(D) \to [0,\infty)$, extend the notation by setting $w(vs) = 0$ when $vs \notin A(D)$. For $v, s \in V(D)$, put*

$$\beta_v^w(s) = \max\Big(\{0\} \cup \{w(us) - w(vs) : u \in V(D),\ v \to u \to s\}\Big).$$

Then there is a vertex $v \in V(D)$ such that

$$\sum_{s\in V(D)} \beta_v^w(s) \ge \sum_{u\in N_1^+(v)} w(vu).$$

The maximum is zero if there is no indicated two-step path. Such paths are determined by the arcs of D, so an arc of weight zero is still present and is not treated as a missing arc.

Proof. Two results of [19] are used as black boxes. This rewrite takes their statements, as described in the source, on trust and does not reprove them.

(iii). Seacrest's Proposition 2 proves that his arc-weighted Conjecture 4, which is assertion (iii) for all nonempty finite oriented graphs and all arc weightings, is equivalent to the unweighted conjecture, which is Theorem 1.1 for all nonempty finite oriented graphs. So Theorem 1.1 gives (iii).

(i) *from* (iii). Set $w(us) = \eta(s)$ on every arc $u \to s$. We claim that for all v, s,

$$\beta_v^w(s) = \begin{cases} \eta(s), & s \in N_2^+(v), \\ 0, & s \notin N_2^+(v). \end{cases}$$

If $s \in N_2^+(v)$, then $w(vs) = 0$ and at least one path $v \to u \to s$ exists; every such path gives the difference $\eta(s) \ge 0$. If $s \in N_1^+(v)$, every eligible difference is $\eta(s) - \eta(s) = 0$. In the remaining cases there is no path $v \to u \to s$ (for $s = v$ by Fact 1.2(d)), so the maximum is 0. Hence the two sides of (iii) become $\eta(N_2^+(v))$ and $\sum_{u\in N_1^+(v)} \eta(u) = \eta(N_1^+(v))$. This is the specialization noted immediately after Seacrest's Proposition 2.

(ii). Let Ψ_D be the $V(D) \times V(D)$ matrix with entry 1 at (v, s) when $s \in N_2^+(v)$, entry -1 when $s \in N_1^+(v)$, and zero otherwise. So $(\Psi_D\eta)_v = \eta(N_2^+(v)) - \eta(N_1^+(v))$. Let $\overleftarrow{D}$ be D with all arcs reversed; it is again a nonempty oriented graph. Then $\Psi_{\overleftarrow{D}} = \Psi_D^{\mathsf{T}}$: s is an out-neighbor of v in $\overleftarrow{D}$ iff $v \in N_{1,D}^+(s)$, and s is at exact distance two from v in $\overleftarrow{D}$ iff there is a path $s \to u \to v$ in D with $v \ne s$ and $s \not\to v$, i.e. iff $v \in N_{2,D}^+(s)$. Seacrest's Theorem 1 applies Farkas' lemma to give either a nonzero $\eta \ge 0$ with $\Psi_D\eta \ge 0$, or a $\xi \ge 0$ with $\Psi_D^{\mathsf{T}}\xi < 0$ at every coordinate. The latter says $\xi(N_2^+(v)) < \xi(N_1^+(v))$ for every v in $\overleftarrow{D}$, which contradicts (i) applied to $\overleftarrow{D}$ and ξ. So the former holds; dividing η by its total weight proves (ii). □

8 Directed-cycle consequences

In an oriented graph with no directed triangle, every exact second out-neighbor is a non-neighbor. Combining this observation with Theorem 1.1 gives degree and order bounds. The first assertion below is the oriented-graph form of Thomassé's non-neighbor statement recorded by Sullivan [21, Conjecture 6.12]. The consequence using both minimum indegree and minimum outdegree is also noted in [21, Section 3]. The regular case gives the girth-four lower bound in the Behzad–Chartrand–Wall minimum-order conjecture [1], also recorded by Sullivan [21, Section 4].

Corollary 8.1 (Non-neighbors and directed cycles)**.** *Let D be a nonempty finite oriented graph of order n. For $v \in V(D)$, write*

$$N_1^-(v) = \{u : u \to v\}, \qquad d^{\pm}(v) = |N_1^{\pm}(v)|, \qquad \delta^{\pm}(D) = \min_{x \in V(D)} d^{\pm}(x).$$

Define the non-neighbors of v, excluding v itself, by

$$N^0(v) = V(D) \setminus \big(\{v\} \cup N_1^+(v) \cup N_1^-(v)\big).$$

Then the following hold.

(i) *If D has no directed 3-cycle, then some vertex v satisfies*

$$d^+(v) \le |N^0(v)|, \qquad n - 1 \ge 2d^+(v) + d^-(v).$$

(ii) *If $\delta^+(D) \ge n/3$ and $\delta^-(D) \ge n/3$, then D contains a directed 3-cycle.*

(iii) *If $r \ge 1$ is an integer, $d^+(v) = d^-(v) = r$ for every vertex, and D has directed girth exactly four, then $n \ge 3r + 1$.*

Here directed girth is the length of a shortest directed cycle.

Proof. (i). Suppose D has no directed 3-cycle. We first show $N_2^+(v) \subseteq N^0(v)$ for every v. Let $w \in N_2^+(v)$ and choose a path $v \to u \to w$. By definition $w \ne v$ and $w \notin N_1^+(v)$. If $w \in N_1^-(v)$, then $v \to u \to w \to v$ would be a directed 3-cycle. So $w \in N^0(v)$. Theorem 1.1 gives a vertex v with

$$d^+(v) \le |N_2^+(v)| \le |N^0(v)|.$$

Since D is oriented, $N_1^+(v)$ and $N_1^-(v)$ are disjoint, and neither contains v. Hence $V(D) \setminus \{v\}$ is the disjoint union of $N_1^+(v)$, $N_1^-(v)$, $N^0(v)$, and

$$n - 1 = d^+(v) + d^-(v) + |N^0(v)| \ge 2d^+(v) + d^-(v).$$

(ii). If D had no directed 3-cycle, the vertex v of (i) would give

$$n - 1 \ge 2d^+(v) + d^-(v) \ge 2\delta^+(D) + \delta^-(D) \ge n,$$

a contradiction. Since degrees are integers, the hypothesis is $\delta^{\pm}(D) \ge \lceil n/3 \rceil$, which includes the endpoint when 3 divides n.

(iii). Directed girth four excludes directed 3-cycles, so (i) applies, and with $d^+(v) = d^-(v) = r$ it reads $n - 1 \ge 3r$. □

9 Verification checklist

This rewrite checked every step of Sections 4–6 and of the two corollary proofs and found none it could not justify, so there are no "unverified" marks. The only statements taken on trust are the two cited results of [19] used in Corollary 7.1. A sceptical reader should check the following first.

1. **The inclusion** (5) **and its two uses** (Lemma 4.2, Step 5(d)–(f)). It forces every newly gained second out-neighbor into $Q \cap F(T)$ and every vertex of C to be lost. It depends on deleting *all* arcs from S to $E \setminus T$.

2. **The exact identity** $g(T \cup Q) - g(T) = |C| - |Q \cap F(T)|$ (Lemma 4.2, Step 5(h)), and that the iteration in Step 6 always restarts from the original D.

3. **The second-image bound and the algebra of the blowup** (Lemma 4.3, Steps 5–6), in particular $|H_0| - 2|F(S)| + |S \cap F(S)| = |F^2(S) \setminus F(S)| - |F(S) \setminus S|$, and the separate case $S = V(D)$ (Step 8).

4. **The cover-to-pruning count** (10) (Lemma 5.2, Step 2). Each covered point of Z forces one of its copies into K; this is where the "uncovered after pruning" term arises.

5. **Maximality of α and β in their own subgraphs** (Lemma 5.2, Step 4), which uses the choice of $\mathcal{M}$ with fewest (E1) edges.

6. $NL = AB$ **and the rank chain** (13) (Steps 5–6).

7. **The simultaneous choice of coefficients** (Step 7). The variables are shared between blocks and between the matrices L, A and $N, B, \Phi^{p,i}$, so the argument must go through one nonzero product polynomial; within each block distinct positions carry distinct variables.

8. **The kernel restriction** (15) (Step 9), especially the rows with $(p, s') \notin H$ and columns with $(i, j') \notin Z$, where the required equation holds only because all coefficients vanish; then the independence of the Bu^r (Step 10).

9. **The hypotheses fed to the pruning lemma in Proposition 6.1**: diagonal members are conflict-free (Step 7), H consists of old uncovered points so $|H| \le d$ (Step 8), and the final arithmetic $M + 2d - (d + d') + d' = M + d$ (Step 9). Also check the one use of orientation, compatibility of (Δ, Δ) (Step 2), and that columns and rows are proper (Step 3) so that (18) applies.

References

[1] Mehdi Behzad, Gary Chartrand, and Curtiss E. Wall. On minimal regular digraphs with given girth. *Fundamenta Mathematicae*, 69(3):227–231, 1970. doi.

[2] Fábio Botler, Phablo F. S. Moura, and Tássio Naia. Seymour's second neighborhood conjecture for orientations of (pseudo)random graphs. *Discrete Mathematics*, 346(12):113583, 2023.

[3] James Brantner, Greg Brockman, Bill Kay, and Emma Snively. Contributions to Seymour's second neighborhood conjecture. *Involve*, 2(4):387–395, 2009.

[4] Jake Brukhman. A dense-case theorem for Seymour's second neighborhood conjecture, 2026. arXiv:2608.11530v1, August 12, 2026.

[5] Guantao Chen, Jian Shen, and Raphael Yuster. Second neighborhood via first neighborhood in digraphs. *Annals of Combinatorics*, 7(1):15–20, 2003.

[6] Moussa Daamouch, Darine Al-Mniny, and Salman Ghazal. About the second neighborhood conjecture for tournaments missing two stars or disjoint paths. *Contributions to Discrete Mathematics*, 20(2):363–383, 2025.

[7] Suresh Dara, Mathew C. Francis, Dalu Jacob, and N. Narayanan. Extending some results on the second neighborhood conjecture. *Discrete Applied Mathematics*, 311:1–17, 2022.

[8] Nathaniel Dean and Brenda J. Latka. Squaring the tournament—an open problem. *Congressus Numerantium*, 109:73–80, 1995.

[9] Jack Edmonds. Systems of distinct representatives and linear algebra. *Journal of Research of the National Bureau of Standards, Section B*, 71B(4):241–245, 1967.

[10] Alberto Espuny Díaz, António Girão, Bertille Granet, and Gal Kronenberg. Seymour's second neighbourhood conjecture: random graphs and reductions. *Random Structures & Algorithms*, 66(1):e21251, 2025.

[11] Dror Fidler and Raphael Yuster. Remarks on the second neighborhood problem. *Journal of Graph Theory*, 55(3):208–220, 2007.

[12] David C. Fisher. Squaring a tournament: A proof of Dean's conjecture. *Journal of Graph Theory*, 23(1):43–48, 1996. doi.

[13] Salman Ghazal. Seymour's second neighborhood conjecture for tournaments missing a generalized star. *Journal of Graph Theory*, 71(1):89–94, 2012.

[14] Frédéric Havet and Stéphan Thomassé. Median orders of tournaments: A tool for the second neighborhood problem and Sumner's conjecture. *Journal of Graph Theory*, 35(4):244–256, 2000. doi.

[15] Hao Huang and Fei Peng. An improved bound on Seymour's second neighborhood conjecture, 2024. arXiv:2412.20234v1, December 28, 2024.

[16] Yoshihiro Kaneko and Stephen C. Locke. The minimum degree approach for Paul Seymour's distance 2 conjecture. *Congressus Numerantium*, 148:201–206, 2001.

[17] Dénes Kőnig. Graphok és matrixok. *Matematikai és Fizikai Lapok*, 38:116–119, 1931. In Hungarian.

[18] Arpan Sadhukhan, R. B. Sandeep, and Sagnik Sen. A proof of Seymour's second neighborhood conjecture for oriented graphs with minimum out-degree equal to 7, 2026. arXiv:2606.30588v1, June 29, 2026.

[19] Tyler Seacrest. The arc-weighted version of the second neighborhood conjecture. *Journal of Graph Theory*, 78(3):219–228, 2015. Preprint (numbering used here): arXiv:1212.1883v1.

[20] Tyler Seacrest. Seymour's second neighborhood conjecture for subsets of vertices, 2019. arXiv:1808.06293v3 [math.CO], revised 12 April 2019.

[21] Blair D. Sullivan. A summary of results and problems related to the Caccetta–Häggkvist conjecture. American Institute of Mathematics, 2006. AIM workshop report dated April 14, 2006; report; Section 3 and Conjecture 6.12.